

Transformation in ETL – Assignment

Name: Sumit Mishra

Course: Data Analytics / ETL Concepts

Question 1: Define Data Transformation in ETL and explain why it is important.

Data Transformation is the process of converting raw data into a clean, structured, and suitable format before loading it into a data warehouse or analytical system. In the ETL process (Extract, Transform, Load), data is first extracted from different sources, then transformed through cleaning, formatting, and standardization, and finally loaded into a target system such as a database or data warehouse.

Importance of Data Transformation:

- Improves data quality by removing errors and duplicates.
- Standardizes data formats coming from multiple sources.
- Makes data suitable for analysis and reporting.
- Helps integrate data from different systems.
- Enables better decision-making using accurate data.

Question 2: List four common activities involved in Data Cleaning.

1. Handling Missing Values – Filling missing values using mean, median, mode, or removing records.
2. Removing Duplicate Records – Eliminating repeated entries that can distort analysis.
3. Correcting Inconsistent Data – Standardizing values such as Male, male, M into 'Male'.
4. Handling Outliers – Detecting and treating extreme values that may affect analysis.

Question 3: Difference between Normalization and Standardization.

Feature	Normalization	Standardization
Definition	Rescales values between 0 and 1	Transforms data to mean=0 and std=1
Formula	$(X - \text{Min}) / (\text{Max} - \text{Min})$	$(X - \mu) / \sigma$
Range	0 to 1	No fixed range
Use Case	When no major outliers exist	Preferred when outliers exist

Question 4: Techniques to handle missing values in the Age column.

Technique 1: Mean or Median Imputation – Replace missing age values with the average or median value of the column. This method is useful when the number of missing values is small and the data is numerical.

Technique 2: Remove Rows with Missing Values – Records containing missing age values can be removed from the dataset. This method should be used only when the number of missing records is very small and removing them will not significantly impact the dataset.

Question 5: Standardizing Gender Values.

Original Values: [M, male, F, Female, MALE, f]

Standardized Values: [Male, Male, Female, Female, Male, Female]

Question 6: What is One-Hot Encoding?

One-Hot Encoding is a method used to convert categorical variables into a numerical format so that machine learning algorithms can process them. Each category becomes a separate column with binary values (1 or 0).

Color	Red	Blue	Green
Red	1	0	0
Blue	0	1	0
Green	0	0	1

Question 7: Difference between Data Integration and Data Mapping.

Data Integration is the process of combining data from multiple sources into a single unified dataset. Data Mapping is the process of matching fields from the source system to the corresponding fields in the target system.

Question 8: Why Z-score Standardization is preferred when outliers exist.

Min-Max scaling rescales data between 0 and 1 using the minimum and maximum values of the dataset. If an extreme outlier exists, it stretches the range and compresses the remaining values. Z-score standardization uses the mean and standard deviation of the dataset and centers the data around zero. Because it is based on statistical distribution rather than absolute minimum and maximum values, it handles outliers better than Min-Max scaling.